



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO1_At2_Agile Sprint Planning

Case Study Title: Flood Early Warning System

(SDG 13:Climate Action)

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1. Project Vision and Stakeholders

Project Vision

To develop a software system that provides real-time flood warnings, evacuation guidance, and emergency resource coordination to reduce loss of life and property during floods.

Objectives

- Predict flood risk using weather data.
- Send instant flood alerts.
- Display safe evacuation routes.
- Help disaster authorities coordinate rescue operations.

Stakeholders

Stakeholder	Role
Citizens	Receive flood alerts and evacuation guidance
Disaster Management Authority	Monitor flood situations and send alerts
Rescue Teams	Perform rescue operations
Weather Department	Provides rainfall and river-level data
Government	Oversees disaster management

2. Epics and User Stories

Epic 1: User Management

User Story 1

As a citizen, I want to register in the application so that I can receive flood alerts.

User Story 2

As a citizen, I want to log in securely so that I can access my account.

Epic 2: Flood Monitoring

User Story 3

As a disaster management officer, I want to monitor rainfall and river water levels so that I can identify flood risks.

User Story 4

As a citizen, I want to receive real-time flood alerts so that I can stay safe.

Epic 3: Evacuation Support

User Story 5

As a citizen, I want to view evacuation routes so that I can reach safe locations.

User Story 6

As a citizen, I want to find nearby shelters so that I can stay safe during floods.

Epic 4: Emergency Coordination

User Story 7

As a rescue team member, I want to receive emergency requests so that I can respond quickly.

User Story 8

As a disaster management officer, I want to manage emergency resources so that rescue operations are efficient.

3. Acceptance Criteria

User Story 1

Given a new user

When the user enters valid details

Then the account is created successfully.

User Story 4

Given flood risk is detected

When the prediction system identifies danger

Then an alert is sent immediately to all registered users.

User Story 5

Given a flood alert exists

When the user opens the map

Then safe evacuation routes are displayed.

User Story 6

Given shelters are available

When the user searches nearby shelters

Then the closest shelters are displayed.

4. Product Backlog

Priority User Story

High User Registration

High User Login

High Flood Risk Prediction

High Flood Alert Notification

Medium Evacuation Map

Medium Shelter Information

Low Emergency Resource Management

Low Rescue Team Dashboard

5. Sprint Backlog (Sprint 1)

Selected User Stories

- User Registration

- User Login
- Flood Alert Notification
- Dashboard

Development Tasks

- Design login page
- Develop registration page
- Create dashboard
- Implement flood alert module
- Connect weather data
- Perform testing

6. Story Point Estimation (Fibonacci)

User Story	Story Points	Justification
Registration	3	Simple form
Login	2	Authentication
Dashboard	5	Multiple UI components
Flood Alerts	8	Notification integration
Evacuation Map	8	Map implementation
Shelter Information	5	Database retrieval

7. Sprint Goal

Develop the basic version of the Flood Early Warning System with secure user authentication, flood monitoring dashboard, and real-time flood alert functionality.

Expected Deliverables

- Login Screen
- Registration Screen
- Dashboard
- Flood Alert Module
- Basic Database
- Weather Data Integration
- Tested Prototype

Sprint Duration

2 Weeks

Agile Methodology Used

- Scrum
- Sprint Planning

- Daily Stand-up
- Sprint Review
- Sprint Retrospective

Tools Used

- Figma (UI Design)
- Draw.io / Lucidchart (Diagrams)
- MySQL (Database)
- Java / Python (Backend)
- GitHub (Version Control)

Algorithm Used

Random Forest Algorithm

Reason:

- High prediction accuracy
- Suitable for flood risk prediction
- Handles multiple weather parameters effectively